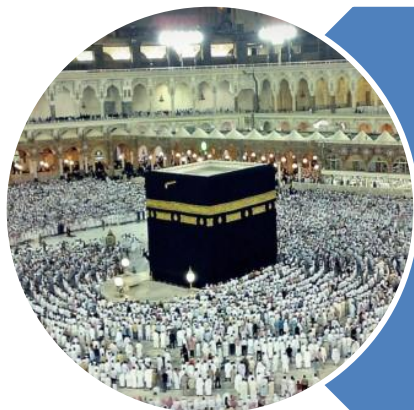


رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي وَاحْلُلْ عُقْدَةً مِنْ لِسَانِي يَفْقَهُوا قَوْلِي



O my Lord! Expand for me my chest [with assurance] and ease for me my task and untie the knot from my tongue that they may understand my speech. **(Quran 20 : 25-28)**

Graph

Introduction

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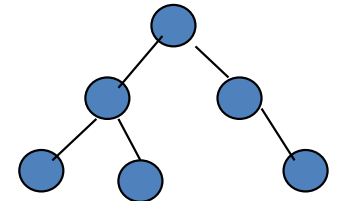
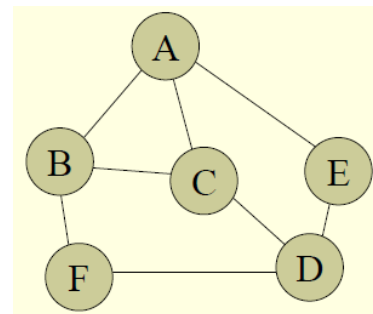
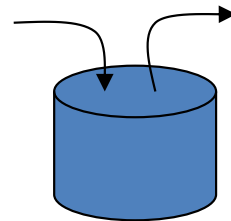
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Graph - Introduction

- Arrays
- Linked List
- Stacks
- Queues
- Tree
- **Graph**



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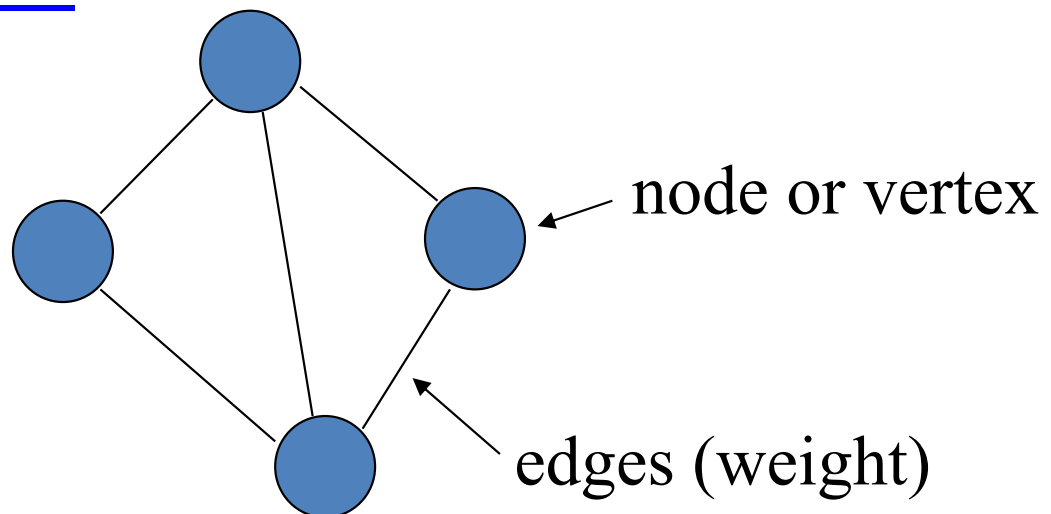
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Graph - Introduction

- Definition
 - A **graph** is a non-linear and non-hierarchical data structure that organizes data elements, called vertices, by connecting them with links, called edges



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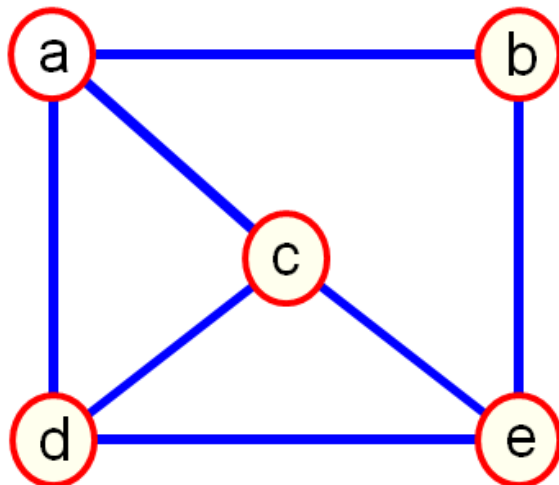
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Graph - Introduction

- A graph $G = (V, E)$ is composed of:
 - V : set of vertices (nodes)
 - E : set of edges (arcs) connecting the vertices V
- An edge $e = (u, v)$ is a pair of vertices

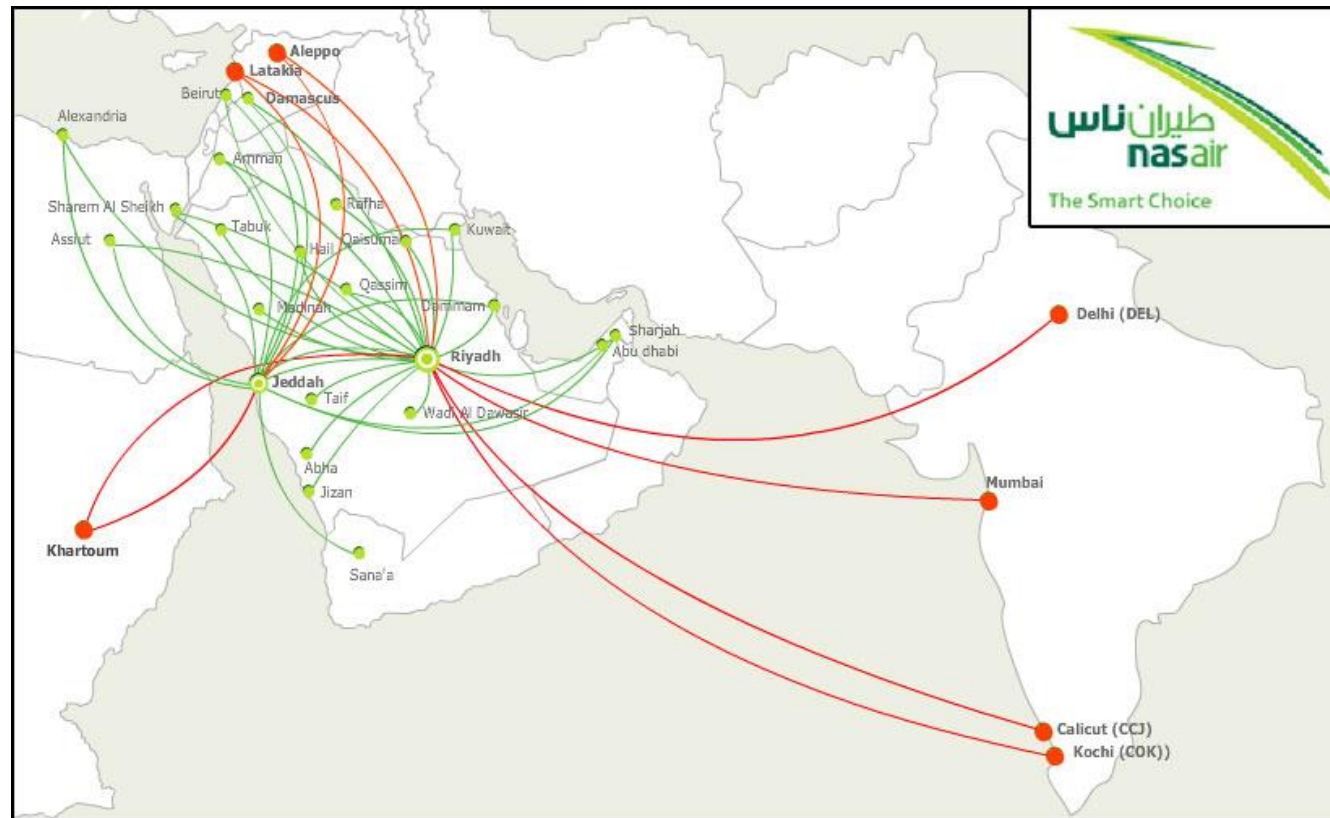


$V = \{a, b, c, d, e\}$

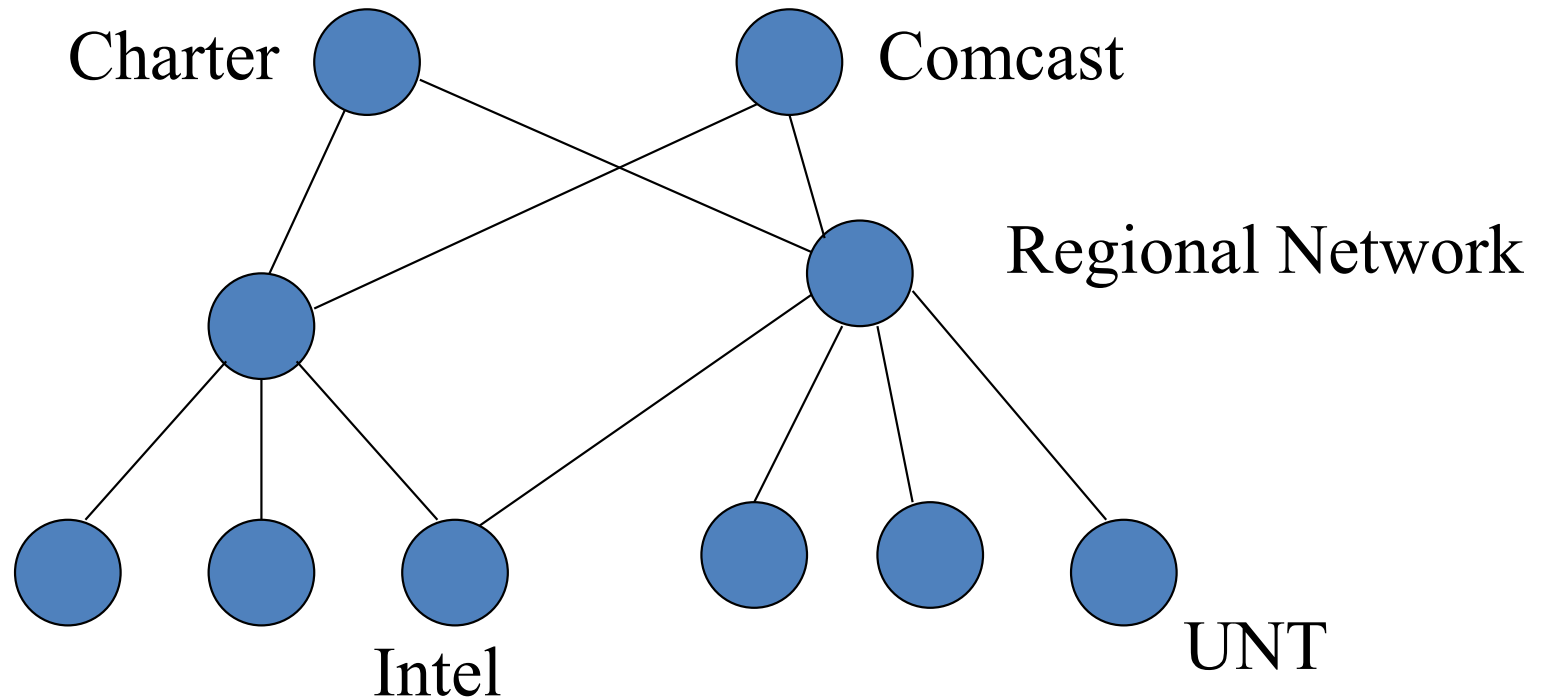
$E =$

$\{(a, b), (a, c), (a, d), (b, e), (c, d), (c, e), (d, e)\}$

Graph Examples - **nasair** flight network

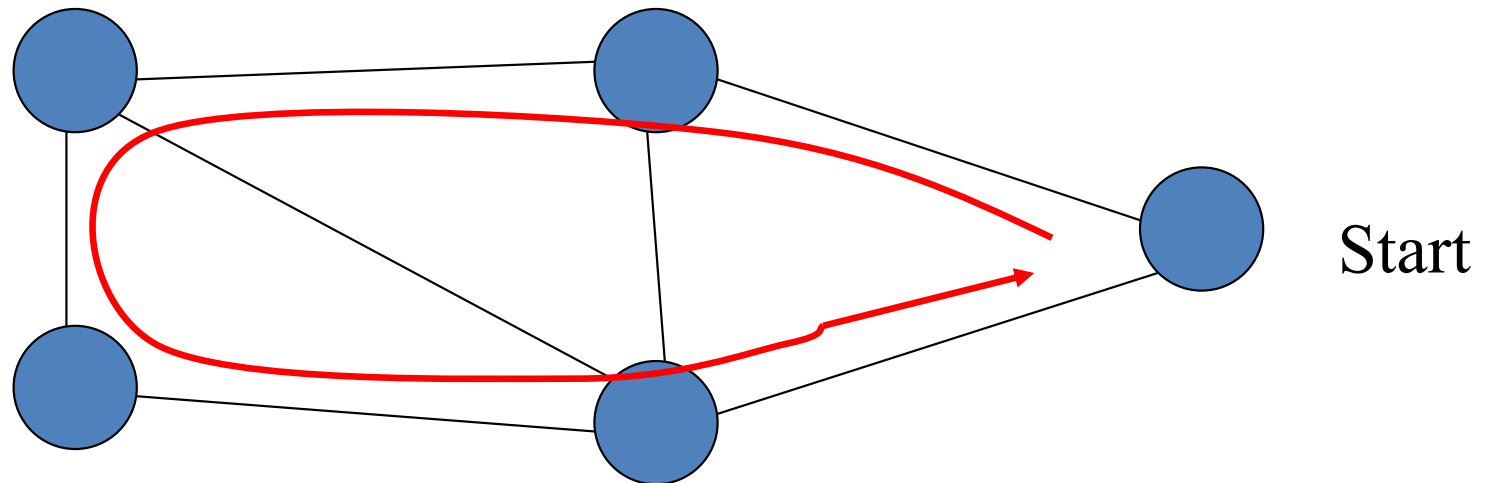


Graph Examples – Computer Network



Graph - Application

- Traveling Salesman

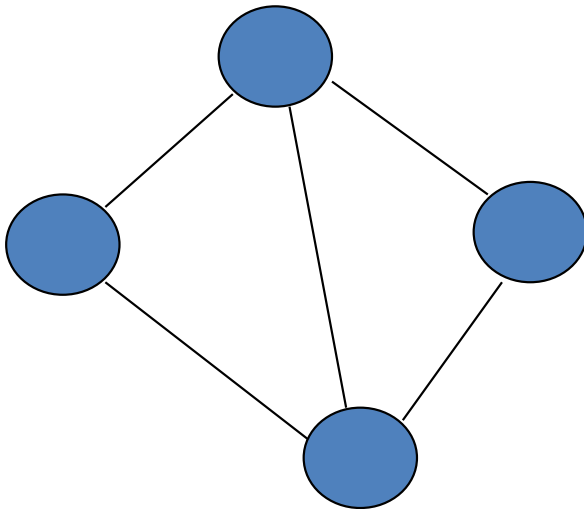


- Find the shortest path that connects all cities without a loop.

Graph - Types

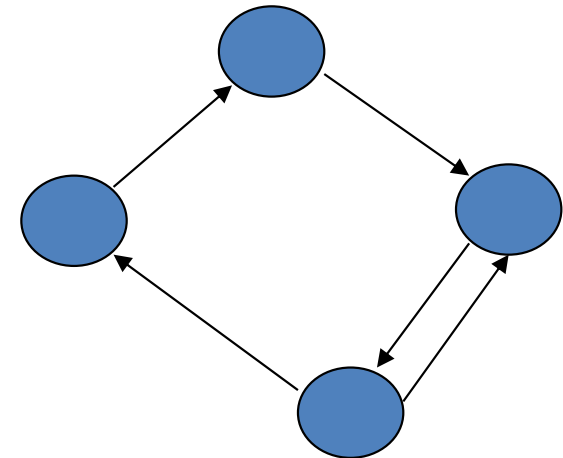
- Undirected Graph

- Edge has no orientation



- Directed Graph

- Edge has oriented vertex

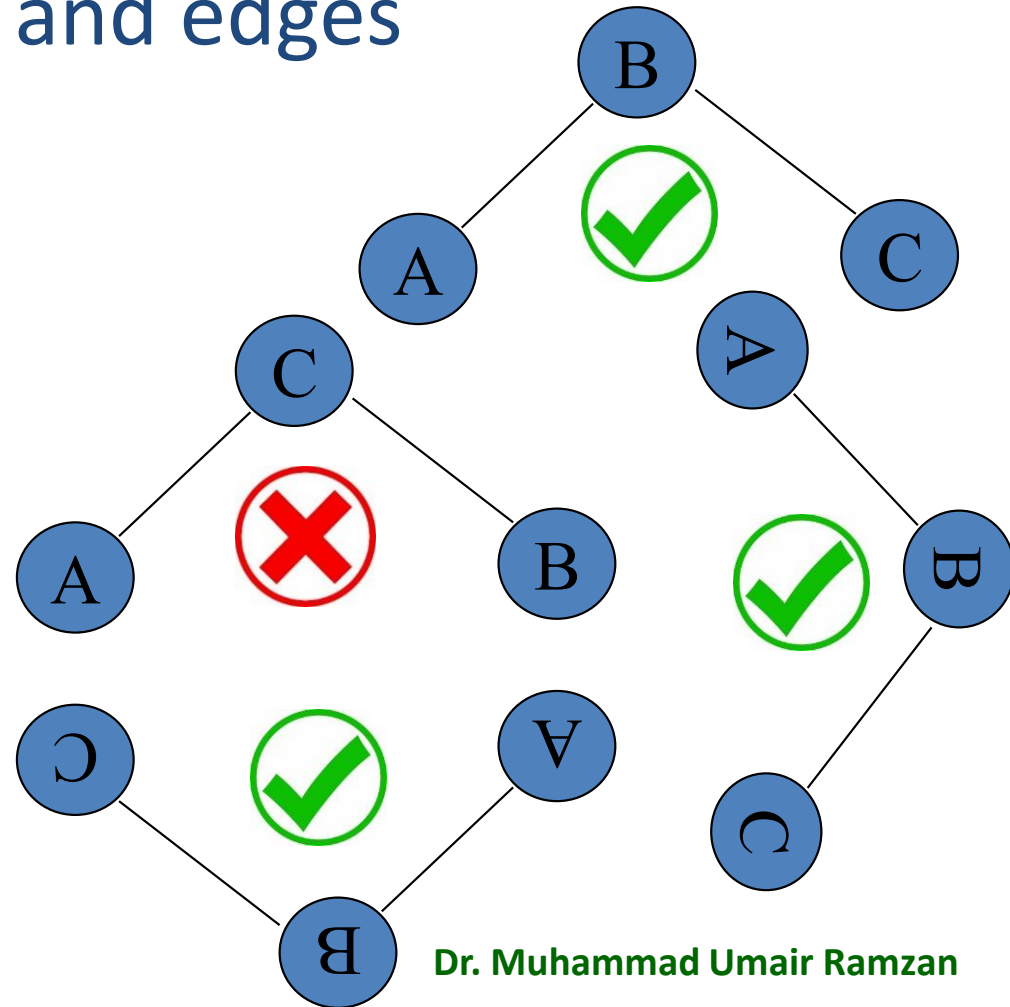
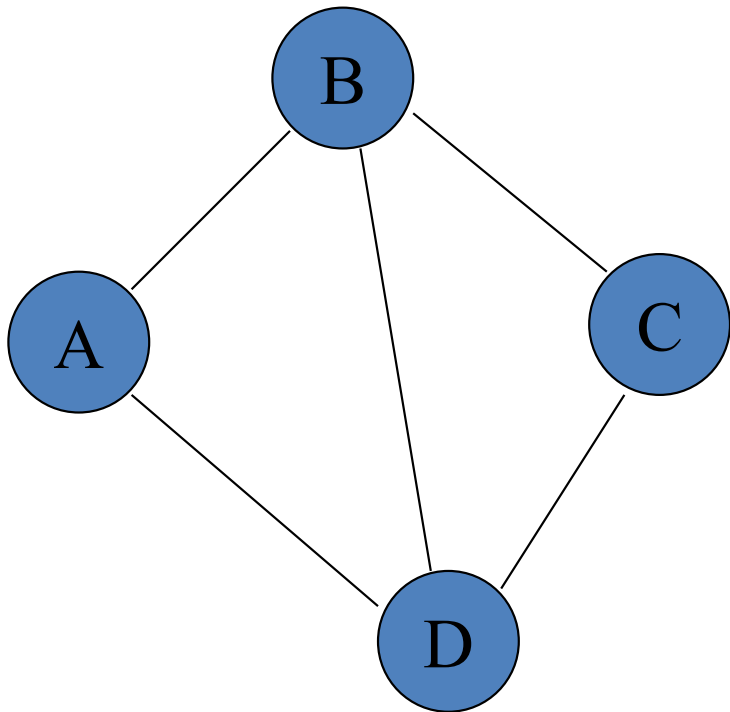


Graph

Terminologies

Terminologies - **Sub graph**

- Subset of vertices and edges



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Terminologies – Path

- A path is a walk from one vertex to any other vertex.
- The vertices traversed to reach to a vertex from an other

○ *ABCD* is a path

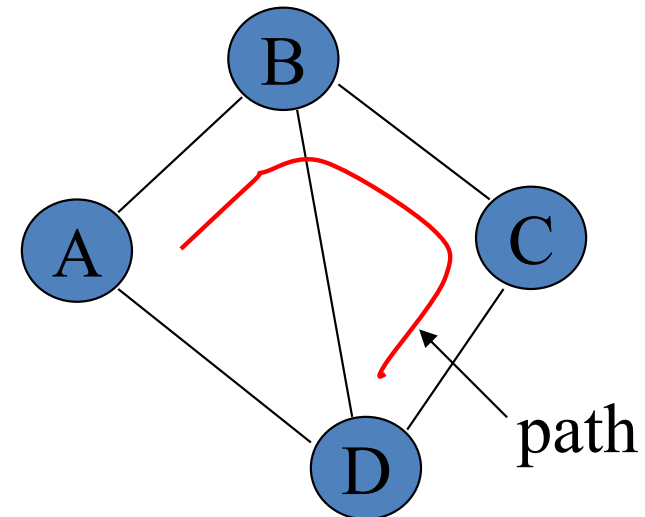
○ ADC ✓

○ ABDA ✓

○ BCDA ✓

○ ABDCBA ✓

○ ABCA ✗



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Terminologies – Simple Path

- A simple path is a path such that all vertices are distinct

○ *ABCD* is a simple path

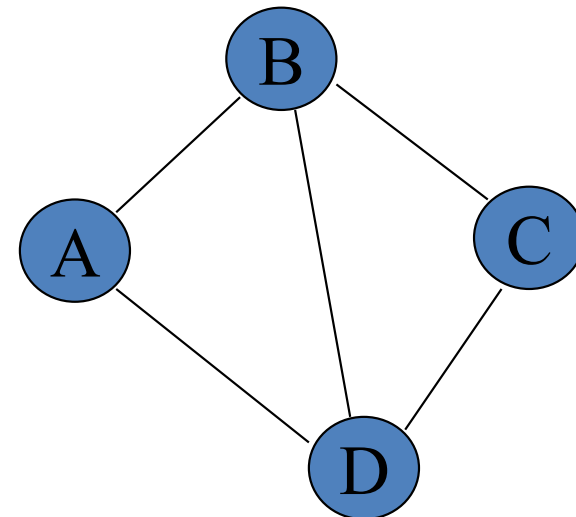
○ ADC 

○ ABDA 

○ BCDA 

○ ABDCA 

○ ABCA 



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Terminologies – Cycle

- A cycle is a path that starts and ends at the same point. For undirected graph, the edges are distinct.

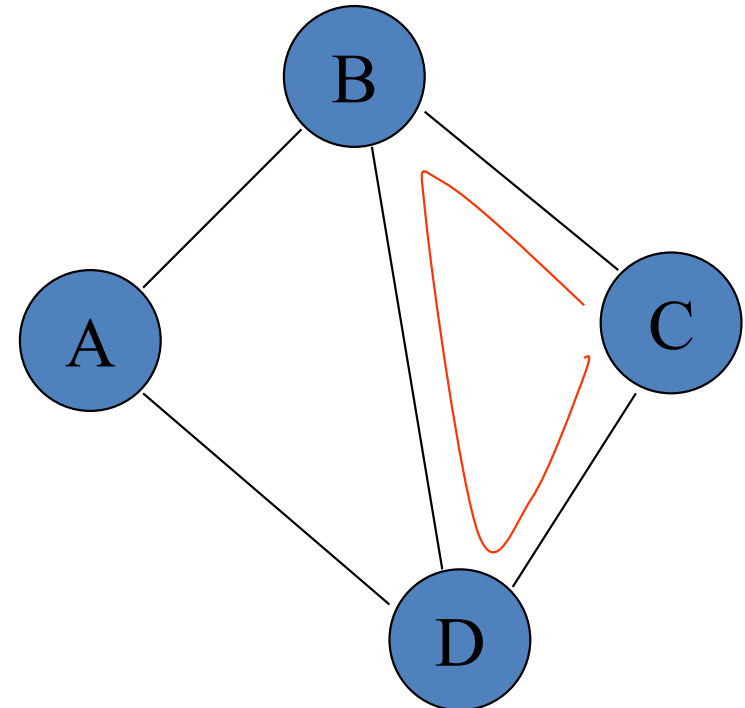
○ *CBDC* is a Cycle

○ ABCD ❌

○ ABDA ✔️

○ BCDB ✔️

○ ABDCBA ✔️



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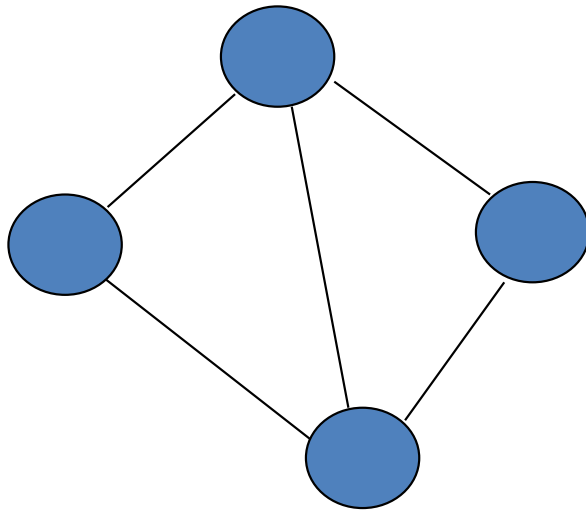
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Graph – Connected / Unconnected

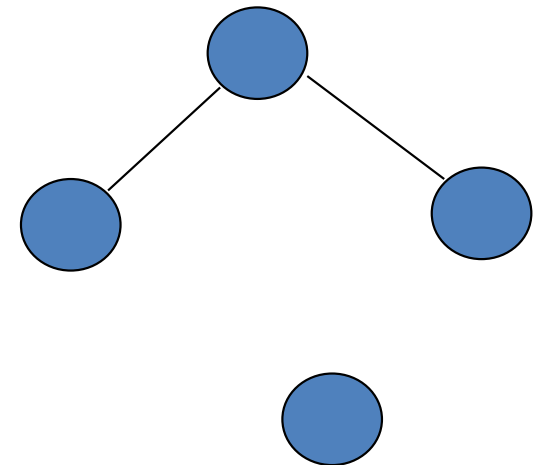
- Connected Graph

- All vertices are connected



- Unconnected Graph

- One or more vertices are not linked



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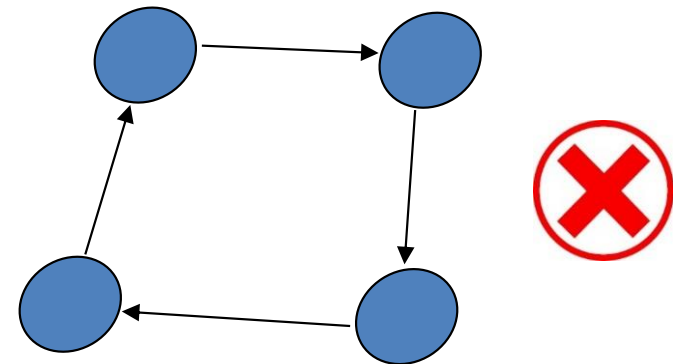
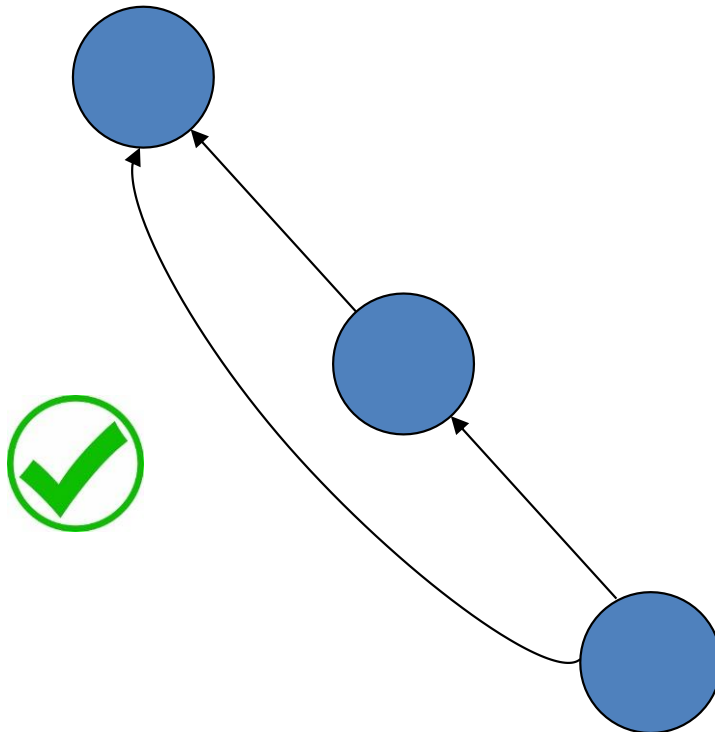
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Terminologies – DAG

- Directed Acyclic Graph (DAG) : directed graph without cycle



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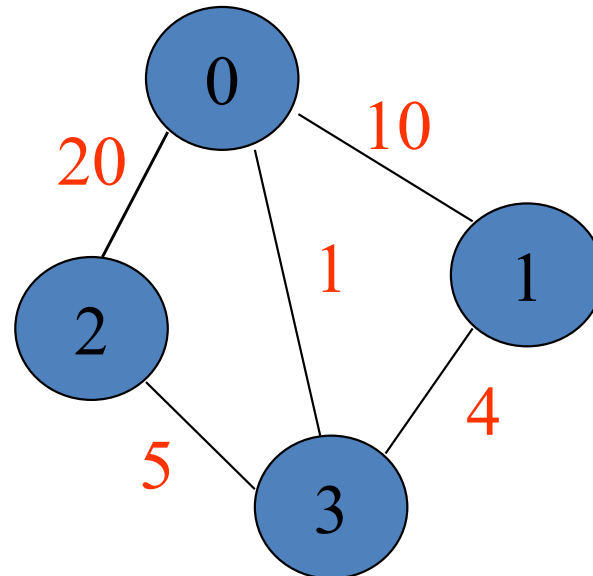
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Terminologies – **Weighted Graph**

- Weighted graph: a graph with numbers assigned to its edges
 - cost, distance, travel time, etc.



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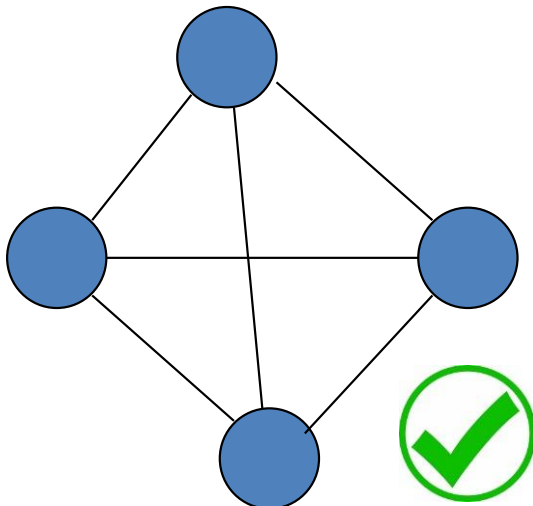
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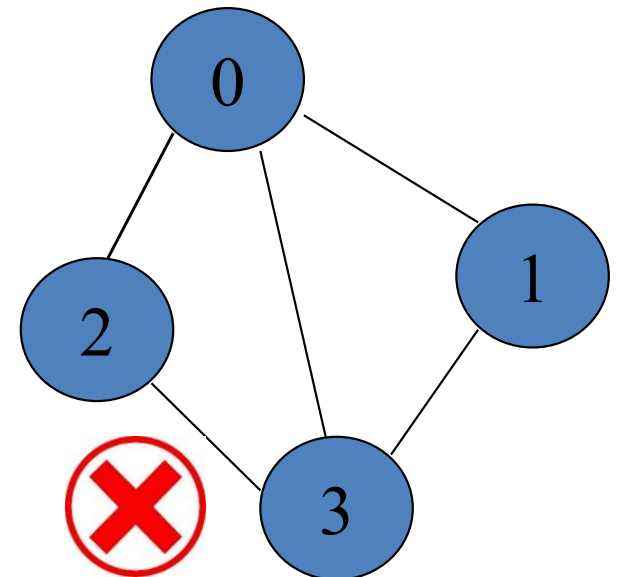
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Terminologies – Complete Graph

- There is a direct edge between every two vertices
- Total edges $E = N(N-1)/2$



Dr. Jonathan (Yahya) Cazalas



Dr. Muhammad Umair Ramzan

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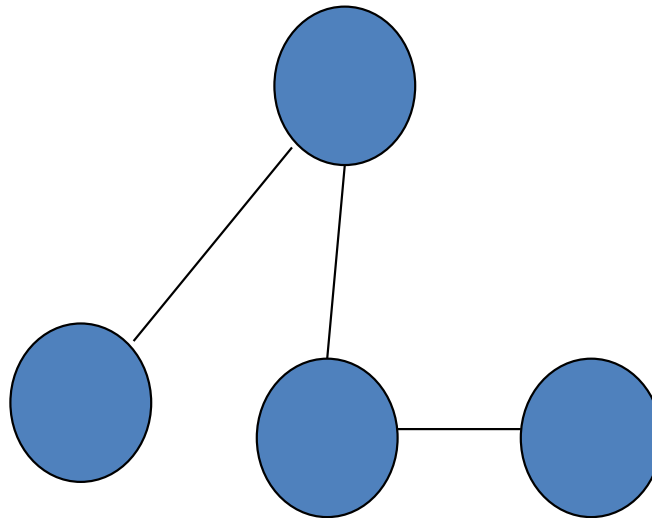
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Terminologies – **Sparse Graph**

- There is a very small number of edges in the graph
- Total edges $E = N - 1$



Graph Representation

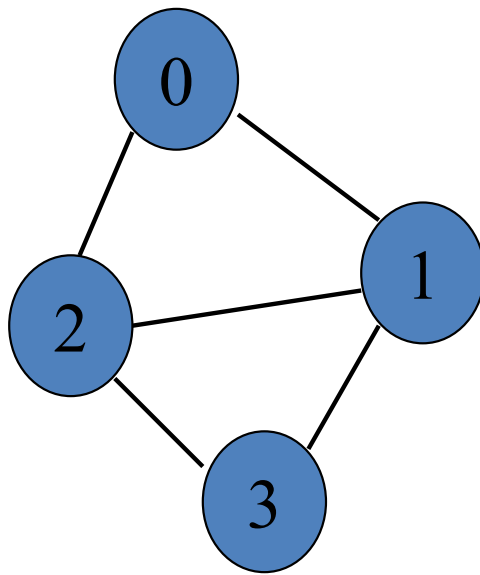
Graph - **Representation**

- Adjacency Matrix
- Adjacency List

Representation – **Adjacency Matrix**

- Assume N nodes in graph
- Use Matrix $A[0...N-1][0...N-1]$
 - if vertex i and vertex j are adjacent in graph, $A[i][j] = 1$,
 - otherwise $A[i][j] = 0$
 - if vertex i has a loop, $A[i][i] = 1$
 - if vertex i has no loop, $A[i][i] = 0$

Representation – Adjacency Matrix



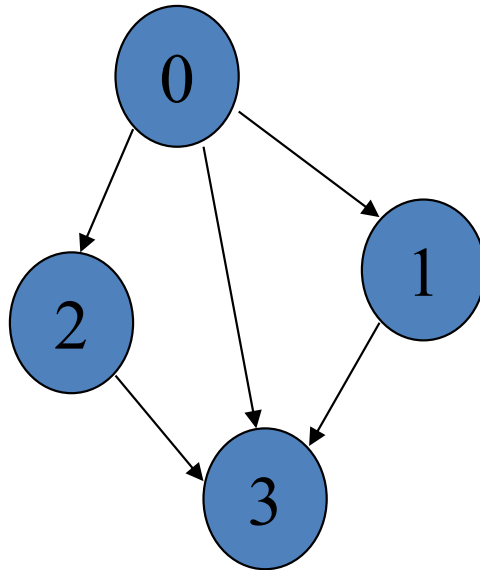
A[i][j]	0	1	2	3
0	0	1	1	0
1	1	0	1	1
2	1	1	0	1
3	0	1	1	0

Undirected Graph

So, Matrix A =

$$\begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

Representation – Adjacency Matrix



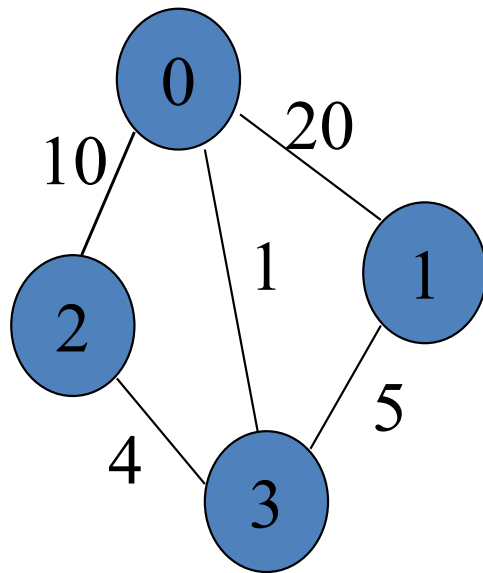
Directed Graph

A[i][j]	0	1	2	3
0	0	1	1	1
1	0	0	0	1
2	0	0	0	1
3	0	0	0	0

So, Matrix A =

$$\begin{pmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Representation – Adjacency Matrix



Weighted Graph

A[i][j]	0	1	2	3
0	0	20	10	1
1	20	0	0	5
2	10	0	0	4
3	1	5	4	0

So, Matrix A =

$$\begin{pmatrix} 0 & 20 & 10 & 1 \\ 20 & 0 & 0 & 5 \\ 10 & 0 & 0 & 4 \\ 1 & 5 & 4 & 0 \end{pmatrix}$$

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Representation – Adjacency Matrix

- Undirected graph
 - adjacency matrix is symmetric
 - $A[i][j] = A[j][i]$

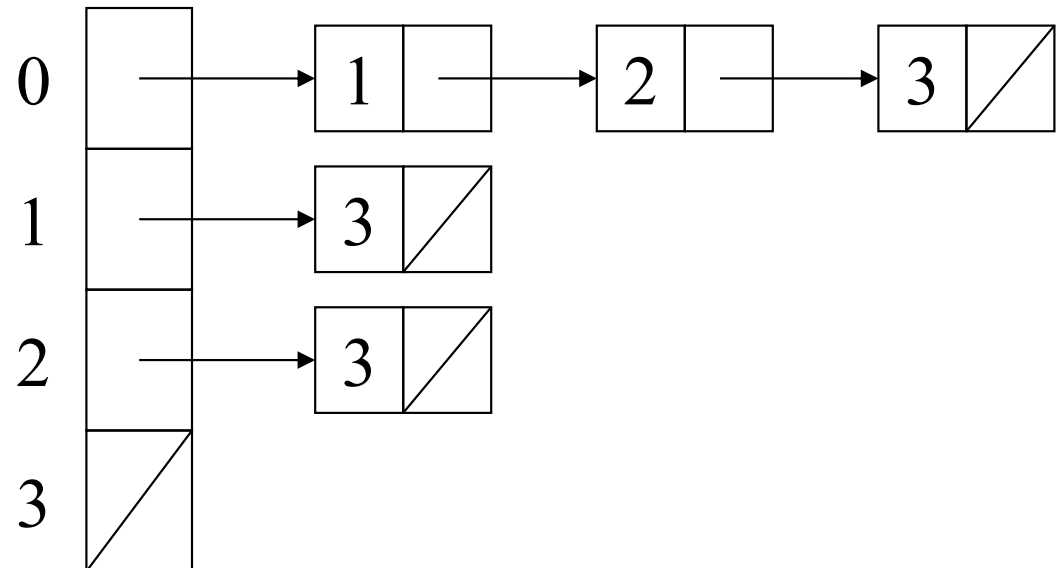
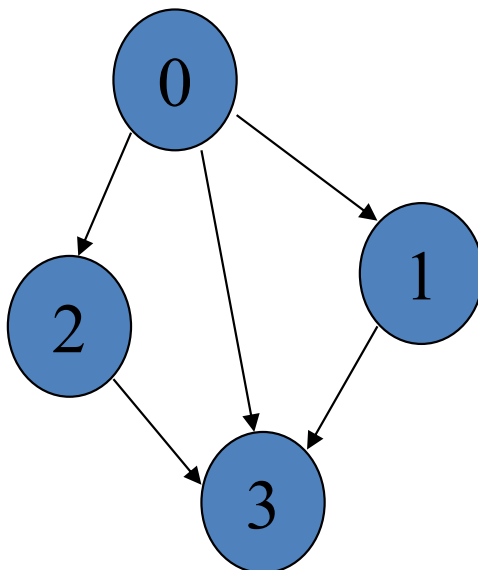
A[i][j]	0	1	2	3
0	0	1	1	0
1	1	0	1	1
2	1	1	0	1
3	0	1	1	0

- Directed graph
 - adjacency matrix may not be symmetric
 - $A[i][j] \neq A[j][i]$

A[i][j]	0	1	2	3
0	0	1	1	1
1	0	0	0	1
2	0	0	0	1
3	0	0	0	0

Representation – Adjacency List

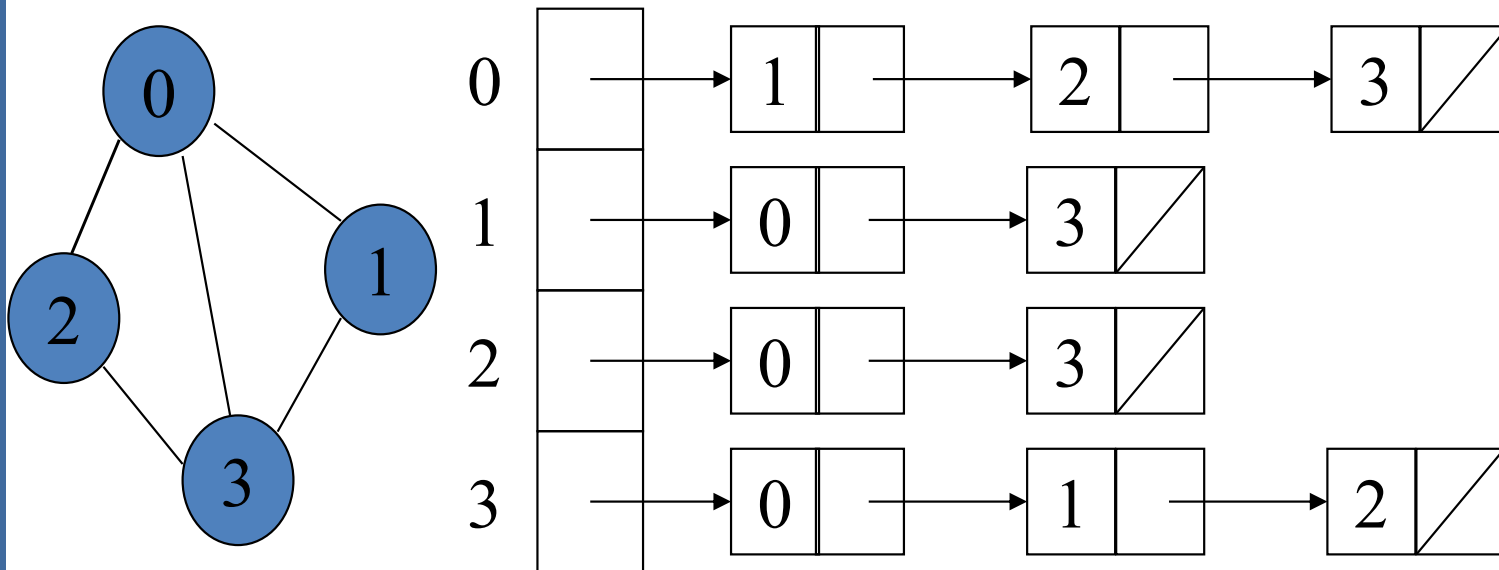
- An array of lists
- The ***i*th** element of the array is a list of vertices that connect to vertex ***i***



Directed Graph

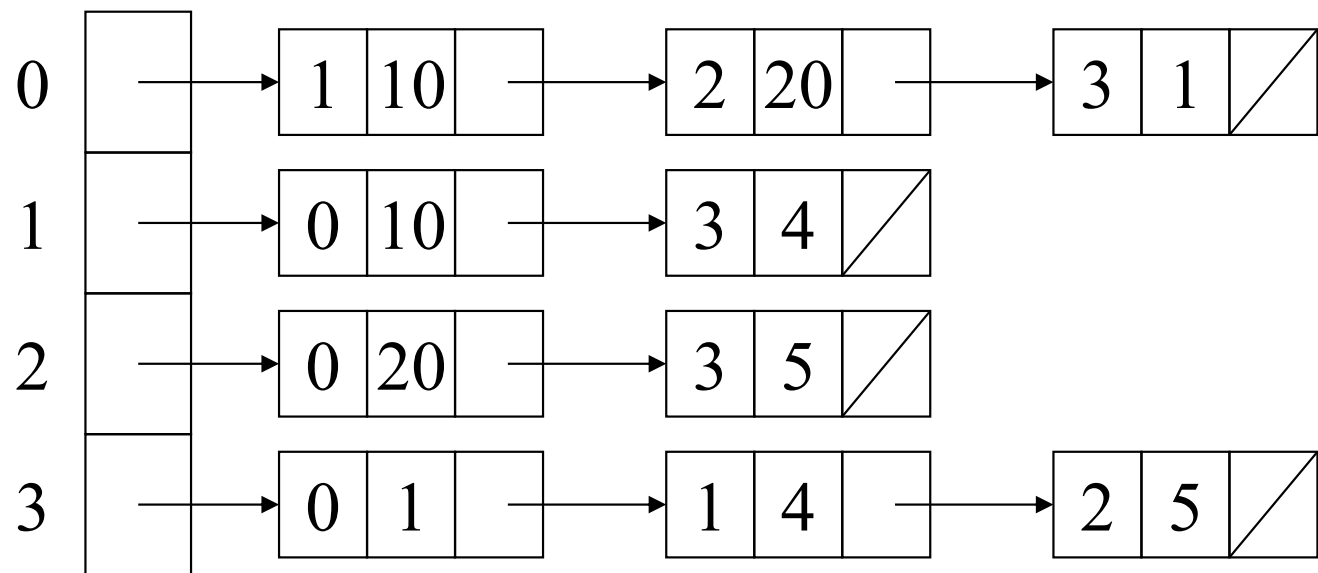
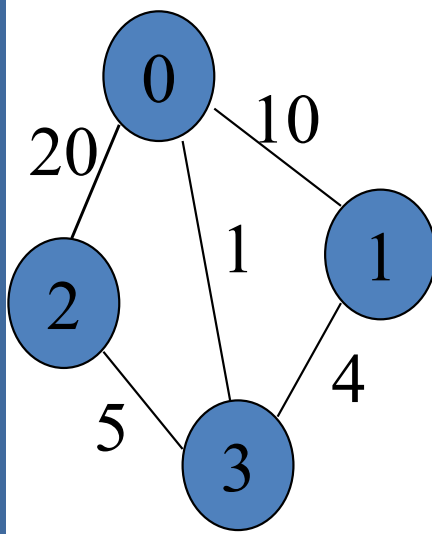
Representation – Adjacency List

- An array of lists
- The *i*th element of the array is a list of vertices that connect to vertex *i*



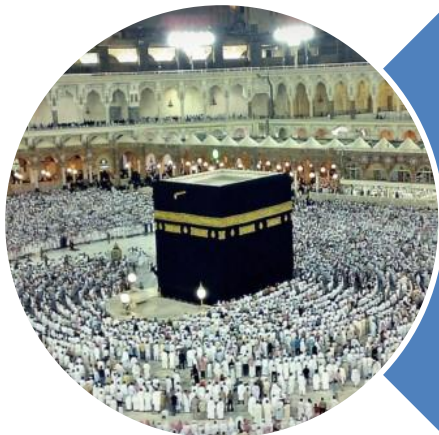
Representation – Adjacency List

- Extend each node with an addition field: weight



Representation – Comparisons

Cost	Adjacency Matrix	Adjacency List
Given two vertices u and v : find out whether u and v are adjacent	$O(1)$	degree of node $O(N)$
Given a vertex u : enumerate all neighbors of u	$O(N)$	degree of node $O(N)$
For all vertices: enumerate all neighbors of each vertex	$O(N^2)$	Summation s of all node degree $O(E)$



Allah (Alone) is Sufficient for us,
and He is the Best Disposer of
affairs (for us). (Quran 3 : 173)